Weather vs Humans: Mumbai

How much of Mumbai's PM2.5 is weather, and how much is us? Station by station, hour by hour.

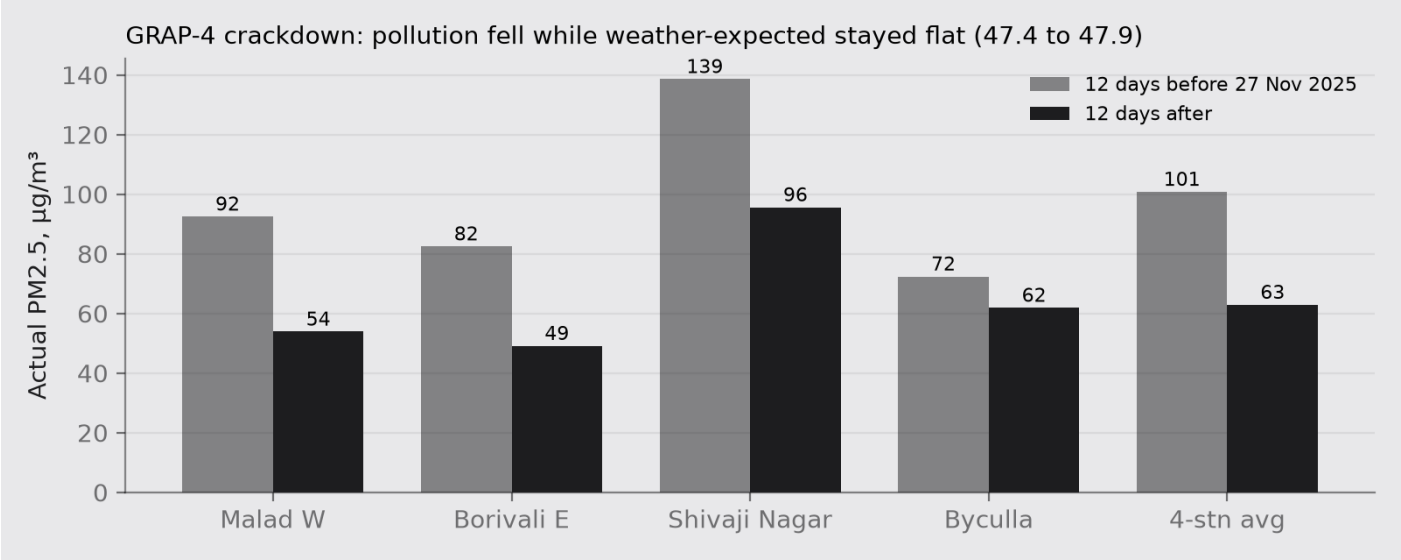
Most-creative submission, Google Mumbai Data Challenge: Air Quality Mapping & Forecasting. Author: Aaryan Singh (BSc Computer Science, RJ College, Mumbai).

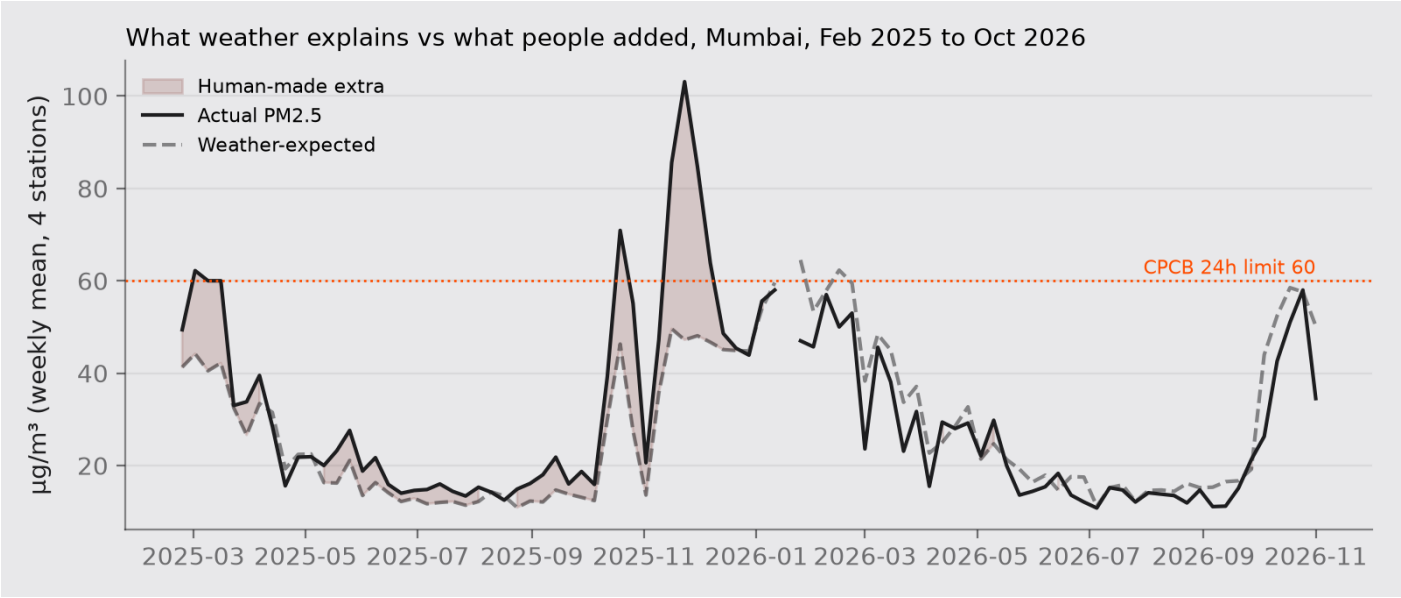
The idea

Bad-air days in Mumbai are usually blamed on "the weather": no wind, a low sea-breeze lid, no rain. That is partly true, which is why it is such a convenient excuse. We trained a model that knows only the weather, the station and the hour. Whatever it cannot explain is the part people added: traffic, construction dust, burning, firecrackers. That gap is the "human meter".

Three findings

- GRAP-4 crackdown worked. BMC enforced GRAP-4 on 27 Nov 2025 (revoked 1 Dec 2025). Across the 4 challenge stations, actual PM2.5 fell from 100.8 to 62.9 ug/m3 (12 days before vs after) while weather-expected stayed at 47.4 to 47.9. In a placebo test, 0 of 358 other winter windows dropped as much.
- Diwali is getting worse, not better. Human extra over +/-2 days: 2019 +7.1, 2024 +23.6, 2025 +44.8 ug/m3/day (peak 124.1 on 21 Oct 2025). That is roughly 2, 5 and 8 cigarette-equivalents per person (Berkeley Earth rule of thumb, approximate).
- Breath debt by neighbourhood (Feb 2025 to Oct 2026): days over the 60 ug/m3 limit that weather alone would NOT have caused. Shivaji Nagar 56 of 61 bad days; Borivali East 50 of 56; Malad West 38 of 46; Byculla 39 of 43. Most bad days are human-made.





Method

- Data: Zindi challenge data 2025-2026 (4 stations), plus public CPCB/MPCB 2018-2020 and 2024 station data for history; hourly weather (wind, boundary-layer height, rain, temperature, humidity) from Open-Meteo ERA5 reanalysis.
- Model: LightGBM on weather, station and hour only (no pollution lags, no calendar), 339,078 hourly rows, leave-one-year-out validation. City daily R2 about 0.57 (0.33 to 0.70 by year).
- Human extra = actual minus weather-expected. Event effects compared with placebo windows to rule out chance.
- Fully reproducible in a free Google Colab notebook; open-source tools only.

Why it matters

- For BMC and MPCB: a weather-adjusted scorecard shows whether an action (GRAP, construction bans, firecracker rules) actually worked, instead of crediting or blaming the wind.
- For citizens: a plain answer to "was today the weather or us?" for their own station.
- For forecasting: the weather-only baseline is a ready feature for the main forecasting task.

What is new

Weather-normalised pollution has been published in research for Indian cities, and source-apportionment tools exist for Mumbai. We did not find a public tool that shows station-level, hourly human-made PM2.5 for Mumbai and tests named events (GRAP, Diwali) against placebo windows.

Honest limits

- R2 of ~0.57 means weather explains only part of the variation; the residual also contains model error, not only human activity.
- Station counts differ by year (10, ~25, 4), so Diwali years are not perfectly comparable.
- The cigarette conversion is an approximation for communication, not a health estimate.
- Diwali 2026 will be added after the final 10 Nov 2026 data update.

Next with Google tools

Gemini to write a plain-language "likely cause" for each spike; Google Maps layer of stations coloured by human share; Looker Studio scorecard for BMC wards.